Aayusha Maharjan

BSc (Hons) Computing, First Class Honours · IoT & Web Development · Technical Author

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PROFILE

First Class BSc (Hons) Computing graduate from The British College, Kathmandu (affiliated with Leeds Beckett University), with practical experience building and deploying IoT systems, full-stack web applications, and technical educational content. Published co-author of a beginner IoT guide affiliated with **IITM Pravartak (India).** I am motivated to pursue postgraduate study to deepen my expertise in computing and contribute to applied research at the intersection of technology, systems design, and human-centred problem-solving.

EDUCATION

BSc (Hons) Computing — First Class Honours *2022 – 2025*

The British College, Kathmandu · Affiliated with Leeds Beckett University, UK

- Relevant modules: Software Engineering, IoT & Embedded Systems, Networks & Security, Database Management, Human-Computer Interaction, Web Development, Project Management
- Final-year capstone project: Real-time IoT Fuel Monitoring System — designed and built end-to-end using Flutter, Firebase, and embedded sensor hardware
- Consistently strong performance across technical and applied modules throughout all three years

A Levels

The British College, Kathmandu

- Subjects: Mathematics, Biology, English, Chemistry

Secondary Education (SEE)

St. Mary's High School, Kathmandu

PUBLICATION

STEAM STARTER: A Complete Guide to Arduino, ESP8266 and ESP32 *Published 2025*

Nepatronix Engineering Solution Pvt. Ltd.

- Co-authored and edited a beginner-level IoT learning guide covering microcontroller fundamentals, Arduino, ESP8266 and ESP32
- Simplified complex embedded systems concepts into structured, accessible content for secondary and post-secondary students across Nepal
- Responsible for technical accuracy, editorial clarity, and consistent pedagogical voice across the full published material

STEAM Booklet Series: Introduction to STEAM (Classes 5–10) *Published 2025*

Nepatronix Engineering Solution Pvt. Ltd. Affiliated with IIT Pravartak, India

– A six-level STEAM booklet series, each volume tailored to the curriculum and cognitive level of its respective class (Classes 5 through 10)

– Simplified foundational concepts in Science, Technology, Engineering, Arts, and Mathematics into structured, age-appropriate content for school students across Nepal

PROFESSIONAL EXPERIENCE

R&D Associate · Nepatronix Engineering Solution Pvt. Ltd. *May 2025 – Present*

- Designed and deployed two production IoT systems: (1) an RFID-based automated attendance tracker and (2) a real-time fuel monitoring solution — leading development from hardware prototype to live deployment
- Built a Flutter mobile application with live Firebase data sync, interactive dashboards, historical trend analysis, and automated alert logic for operational decision support
- Developed Python automation tools that eliminated repetitive workflows, improving team efficiency across multiple internal processes
- Authored STEAM Booklets (Classes 5–10) affiliated with IIT Pravartak — translating embedded systems concepts into curriculum-aligned content used across schools in Nepal
- Maintained structured technical documentation to support knowledge transfer across cross-functional engineering teams

Digital Marketing Intern · Ad Media Pvt. Ltd. *Jun – Dec 2023*

- Scripted and produced content for social media campaigns, translating technical subjects into clear audience-appropriate messaging
- Analysed engagement metrics and user behaviour data to identify trends and improve content strategy and campaign performance
- Collaborated with creative and strategy teams across end-to-end campaign planning, execution, and post-campaign evaluation

KEY PROJECTS

Fuel Monitoring System (Capstone Project) *Flutter · Firebase · ESP32 · IoT Sensors · Mobile App*

Engineered a real-time fuel monitoring system using ESP32 and IoT sensors, integrated with a Flutter mobile application for live data visualization. Implemented fuel analytics, consumption tracking, anomaly detection, and automated alerts to improve transparency and operational efficiency.

RFID Attendance System *RFID Hardware · Mobile App*

Automated attendance tracker integrating RFID card readers with a mobile/desktop interface. Features real-time monitoring, attendance filtering, and export functionality — replacing manual registers across departments.

TrackFlow — Accounting Software *JavaScript · React · Firebase · Tailwind CSS · Vercel*

Full-stack web accounting tool for tracking income, expenses, and financial summaries, with intuitive dashboards designed for accessibility by small businesses and individual users.

Resume Builder *HTML · CSS · JavaScript · AI · Vercel*

Web application enabling users to create, preview, and export professional CVs with real-time formatting and ATS-compatible output. Deployed on Vercel.

Wireless Barcode Scanner Conversion *Embedded Systems · Bluetooth/Wi-Fi*

Converted a wired barcode scanner to wireless operation using embedded Bluetooth/Wi-Fi integration, improving field portability and enabling remote usage.

TECHNICAL SKILLS

Languages	Python · JavaScript · TypeScript · HTML/CSS · PHP · C++ · Java
Frameworks	Flutter · React · Bootstrap
Tools	Firebase · MySQL · Oracle APEX · Git · Selenium · Vercel
Design	Figma · Canva · Microsoft Office
Domains	IoT & Embedded Systems · Full-Stack Web Development · Automation · Technical Writing

ACHIEVEMENTS & RECOGNITION

- First Class Honours — BSc (Hons) Computing, The British College / Leeds Beckett University (2025)
- Published Technical Author — Co-authored STEAM STARTER IoT Guide, affiliated with IIT Pravartak (2025)
- STEAM Curriculum Author — STEAM Booklets Classes 5–10, advancing embedded systems education in schools across Nepal
- MUN Participant — Developed skills in research, structured argumentation, diplomacy, and public speaking

LANGUAGES & ADDITIONAL INFORMATION

Languages	English (Fluent) · Nepali (Native) · Hindi (Conversational)
Availability	Immediate · Open to full-time, part-time, and remote arrangements
References	**Raju Shrestha**, Chief Executive Officer and Founder of Nepatronix. Nepatronix is a leading Nepal-based organization focused on IoT, robotics, and STEM-based technological innovation and education.

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